

## PAPER

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7, 2325Influence of copper and potassium on the  
structure and carbidisation of supported iron  
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This paper addresses the effect of promotion with copper and potassium on the carbidisation kinetics, structure and performance of silica supported iron catalysts for high temperature Fischer–Tropsch synthesis. The promotion with copper and potassium results in the enhancement of iron dispersion. Small hematite nanoparticles were uncovered in the calcined iron catalysts. Activation of the calcined catalysts in carbon monoxide or syngas results in the reduction of hematite to magnetite at 250–300 °C. The magnetite is then carbided into Hägg iron carbide. The promotion with copper and potassium affects the rate of the iron reduction and carbidisation. Copper strongly enhances both hematite reduction and magnetite carbidisation in carbon monoxide and syngas. The presence of potassium hinders reduction of hematite to magnetite. Much higher reaction rates were observed on copper-promoted catalysts than on the potassium-promoted and unpromoted counterparts while the promotion with potassium had a stronger impact on the selectivity.

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## Introduction

Fischer–Tropsch synthesis (FT) is one of the most attractive technologies to convert syngas derived from coal, natural gas, or biomass to liquid fuels and other useful chemicals. It is known that metals of Group VIII such as iron, cobalt and ruthenium are active in FT synthesis. Iron-containing catalysts have several advantages for converting syngas with lower hydrogen concentration. Iron catalysts are active in the water gas-shift reaction and can adjust the H<sub>2</sub>/CO ratio in syngas produced from coal and biomass.<sup>1,2</sup>

Most of the published results on the Fe-containing FT catalysts were obtained on bulk systems. In recent years, however, supported iron catalysts have become a subject of increasing attention. Supported iron catalysts have several advantages compared to their bulk counterparts, such as resistance to attrition, relatively high active surface area and better dispersion of the active phase.<sup>3</sup> The practical use of the supported catalysts is limited by iron–support interactions which may hinder reduction and carbidisation of iron oxides.<sup>4</sup>

The catalytic performance of iron catalysts can be improved by chemical promotion. Both alkali metals (K, Na, etc.) and copper are most common promoters for industrial iron FT catalysts.<sup>3</sup> More recently, de Jong<sup>5,6</sup> *et al.* uncovered a major effect of simultaneous promotion with sodium and sulphur on olefin selectivity in FT synthesis, which was accompanied by low carbon deposition. It is believed that potassium enhances dissociation of carbon monoxide and improves the reaction selectivity to long-chain hydrocarbons.<sup>7–10</sup> The effect of copper on hydrocarbon selectivity in FT synthesis is a subject of multiple controversies. Several earlier works suggest that addition of copper increases the selectivity to C<sub>5+</sub>.<sup>11</sup> Other reports indicate, however, that the selectivity to C<sub>5+</sub> is reduced over copper-promoted catalysts.<sup>12</sup> At the same time, no modification of hydrocarbon selectivity after copper promotion was also observed by several authors.<sup>13</sup> Note that for both copper and potassium, the catalytic performance is also strongly affected by the promoter concentration.

Most of the available experimental evidence suggests that FT synthesis over iron catalysts occurs on iron carbides of different compositions.<sup>14</sup> It is also known that the activity and selectivity of iron catalysts depend on the composition of gas used for iron carbidisation (pure carbon monoxide *versus* syngas with different H<sub>2</sub>/CO ratios). The mechanisms and kinetics of iron carbidisation are therefore of utmost importance for understanding and predicting the catalytic performance of iron catalysts. Our previous report<sup>15</sup> suggests that iron carbide formation from hematite proceeds *via* intermediate formation

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of magnetite. Higher rates of hematite reduction and magnetite carbidisation were observed for smaller supported iron particles than for their larger counterparts. Important information about the structure of iron catalysts during activation and the catalytic reaction can be obtained from *in situ* characterization using for example, X-ray diffraction, X-ray absorption spectroscopy (XANES/EXAFS) or magnetic methods.<sup>16–18</sup>

Catalyst activation can be also affected by promotion. Bartholomew<sup>19</sup> *et al.* investigated the formation of iron carbides in an Fe catalyst supported on silica and promoted by Pt and K. Also it has been established<sup>20</sup> that alkali metals promote carbidisation of iron supposedly by stimulation of CO dissociation on the catalyst surface and acceleration of transformations of Fe<sub>3</sub>O<sub>4</sub> to Fe<sub>3</sub>O<sub>(4-x)</sub>, and ultimately to Fe<sub>x</sub>C.<sup>20–22</sup> Promotion with copper facilitates reduction of iron oxides to the metallic state and formation of bimetallic compounds.<sup>23,24</sup> It should be noted that the effect of copper on iron reduction and carbidisation can be different for the catalysts prepared by impregnation and co-precipitation.<sup>25</sup> Recently Worayingyong *et al.*<sup>26</sup> used time-resolved XANES to investigate the structure of iron catalysts supported by nitrogen-containing carbon nanotubes (NCNT). The presence of Fe<sub>2</sub>O<sub>3</sub> and Fe<sub>3</sub>O<sub>4</sub> phases was uncovered in the potassium promoted NCNT during initial reduction, while Fe, FeO, Fe<sub>2</sub>O<sub>3</sub>, and Fe<sub>3</sub>O<sub>4</sub> phases were detected at a longer reduction time. The NCNT supported iron catalysts promoted with Mn exhibited only the Fe<sub>3</sub>O<sub>4</sub> phase during 60 min of reduction time, and a mixture of FeO and Fe<sub>3</sub>O<sub>4</sub> was observed after a more extended reduction period.

Magnetic methods operating under *in situ* conditions are a powerful tool for characterisation of iron catalysts. These methods can yield information about the concentration of ferri- and ferromagnetic phases, their chemical composition and in some particular cases about nanoparticle sizes. Several *in situ* cells have been designed by the groups of Claeys,<sup>27–29</sup> Schuurman<sup>30</sup> and Chernavskii.<sup>16–18</sup>

The present work addresses the effect of promotion of silica supported iron catalysts with copper and potassium on the carbidisation kinetics in syngas and carbon monoxide, catalyst structure and catalytic performance in high temperature FT synthesis. An *in situ* magnetic method which is particularly suitable to follow the evolution of the ferrimagnetic and ferromagnetic nanoparticles of iron compounds under catalyst activation conditions was used in this work.

## Experimental

### Catalyst preparation

The CARIAC silica gel (Fuji Silysia Chemical, Q-15 grade) with a narrow pore size distribution ( $d_p = 15$  nm) was chosen as a catalytic support. A 15% Fe/SiO<sub>2</sub> catalyst was prepared by conventional incipient wetness impregnation of silica with an aqueous solution of Fe(NO<sub>3</sub>)<sub>3</sub>·9H<sub>2</sub>O of predetermined concentration. A 15% Fe–1% Cu/SiO<sub>2</sub> catalyst was prepared by sequential co-impregnation of silica with iron and copper nitrates (Fe(NO<sub>3</sub>)<sub>3</sub>·9H<sub>2</sub>O + Cu(NO<sub>3</sub>)<sub>2</sub>·6H<sub>2</sub>O), and those of 15%

Fe–1% K/SiO<sub>2</sub> – by co-impregnation of silica with salts of potassium and iron nitrates (KNO<sub>3</sub> + Fe(NO<sub>3</sub>)<sub>3</sub>·9H<sub>2</sub>O) of predetermined concentrations. After each subsequent impregnation the samples were dried in an air flow with an infrared lamp for 2 h and then calcined in an air flow at 450 °C for 4 h in programmable mode with a temperature ramp of 7.5 °C min<sup>–1</sup>. The elemental composition of the obtained catalysts was determined using an ICP-AES 720-ES spectrometer (Agilent Technologies). The results of the elemental analysis are shown in Table 1.

### Catalyst characterisation

**X-ray diffraction.** The X-ray diffraction analysis was performed using a 3-DRON diffractometer with CuK<sub>α</sub> radiation and a graphite monochromator. The measurements were carried out using an angle scanning step of 0.1° and 3 s per point acquisition time. In order to estimate the crystallite size, the acquisition was performed with 0.02° steps and 10 s per point. The crystallite size ( $D$ , nm) was evaluated by the Scherrer formula.

**Mössbauer spectroscopy.** The Mössbauer absorption spectra were measured using a Perseus spectrometer (Russia) working at constant velocities and were analysed using the Univem MS program (Rostov-on-Don, Russia) based on a nonlinear least squares fitting routine. A standard  $\gamma$ -source of <sup>57</sup>Co in a metallic Rh matrix with an activity of 20 mCi (a product of Ritverc, Russia) was employed. Isomer shifts in this paper are presented relative to  $\alpha$ -Fe at 25 °C.

***In situ* magnetic method.** The magnetic measurements were performed *in situ* in a vibrating sample magnetometer. Samples weighing 10 mg were placed in a measuring cell of 0.3 cm<sup>3</sup> volume<sup>16–18</sup> which served also as a plug-flow micro-reactor. The sample was fixed between the two membranes of porous silica. In a non-isothermal experiment, the sample was heated to a predetermined temperature in a flow of CO (or synthesis gas). The gas flow rate in all experiments was 10 cm<sup>3</sup> min<sup>–1</sup>. The gases were passed first through a column with  $\gamma$ -Al<sub>2</sub>O<sub>3</sub> heated to 300 °C for the removal of iron carbonyl impurities. The temperature ramping was performed at 10 °C min<sup>–1</sup> to a predetermined temperature, while the magnetization was recorded continuously at 1 Hz. After reaching the desired temperature, the sample was cooled under a flow of CO (or syngas) at 10 °C min<sup>–1</sup>; the temperature dependence of magnetization was then recorded. In an isothermal experiment, the sample was heated to a predetermined temperature

**Table 1** Catalyst elemental compositions and  $\alpha$ -Fe<sub>2</sub>O<sub>3</sub> crystallite sizes calculated by XRD

| Catalyst               | Elemental composition, wt% |           |             | $D_{\text{Fe}_2\text{O}_3}$ from (104), nm |
|------------------------|----------------------------|-----------|-------------|--------------------------------------------|
|                        | Fe                         | Cu        | K           |                                            |
| Fe/SiO <sub>2</sub>    | 12.1 ± 1.2                 | —         | —           | 20                                         |
| FeCu/SiO <sub>2</sub>  | 12.5 ± 1.2                 | 1.2 ± 0.1 | —           | 18                                         |
| FeK/SiO <sub>2</sub>   | 12.5 ± 1.2                 | —         | 0.75 ± 0.07 | 17                                         |
| FeCuK/SiO <sub>2</sub> | 12.4 ± 1.2                 | 1.3 ± 0.1 | 0.71 ± 0.07 | 15                                         |

in argon, then the argon flow was switched to that of CO (or syngas), after which the magnetization was recorded. The CO<sub>2</sub> concentration in the reactor outlet was continuously recorded using an infrared detector.<sup>18</sup>

After carbidisation and cooling to room temperature, the thermal dependencies of magnetization were measured in a flow of argon. The Curie temperature was determined from these curves with a high accuracy. The Curie temperature is a characteristic feature of different iron ferri- and ferromagnetic phases. Several inflection points in the thermomagnetic curves indicate the presence of several magnetic phases. Analysis of thermomagnetic curves in case of two magnetic phases, for which the specific value of the magnetization is known, provides information about their relative fraction in the catalyst.

**H<sub>2</sub> temperature-programmed reduction.** The TPR experiments were performed in two setups: in a standard TPR setup with a quartz reactor and in a similar microreactor which was at the same time a measuring cell of the vibrating sample magnetometer. The vibrating sample magnetometer enabled continuous recording of the magnetization during the reduction. In the conventional TPR setup, the maximum temperature was 800 °C, while in the magnetometer the temperature did not exceed 600 °C.

The change in the concentration of hydrogen at the reactor outlet was recorded using a thermal conductivity detector. The TPR experiment parameters (flow rate, heating rate, catalyst weight) were adjusted according to the relation:

$$\frac{\beta S}{fC} < 20$$

where  $\beta$  is the heating rate (°C s<sup>-1</sup>),  $S$  is the amount of the reduced oxide (mol),  $f$  is the flow rate (cm<sup>3</sup> s<sup>-1</sup>), and  $C$  is the concentration of hydrogen in the gas mixture at the reactor inlet.

These conditions provide the best resolution of the peaks in the TPR profile.<sup>31</sup> The heating rate is generally in the range of 0.2 °C s<sup>-1</sup> at the reduction gas flow of 0.5 cm<sup>3</sup> s<sup>-1</sup> for the catalyst loading of 0.02 g.

**FT catalytic tests.** The catalytic performance in FT synthesis was studied in a high-pressure tubular flow fixed-bed reactor. The catalyst with a grain diameter of 90 µm diluted in silicon carbide (1 g of catalyst per 5 g of SiC) was packed into a stainless steel reactor (i.d. of 8.0 mm) and activated in flowing carbon monoxide at 350 °C for 16 h. After the activation, the catalysts were cooled down to 160 °C and a flow of premixed syngas (H<sub>2</sub>/CO = 2) was gradually introduced through the catalyst. When the pressure attained 20 bar, the temperature was slowly increased to 300 °C.

The gaseous reaction products were analysed by online gas chromatography. Analysis of permanent gases was performed using a molecular sieve column and a thermal conductivity detector. The reactant and stream products were analysed on-line using a GC-456 Bruker gas chromatograph equipped with thermal conductivity and flame ionization detectors. H<sub>2</sub>, N<sub>2</sub>, CO, CH<sub>4</sub> and CO<sub>2</sub> were separated with a CTR-1 column and analysed by a TCD. The C<sub>1</sub>–C<sub>5</sub> hydrocarbons

and light alcohols were separated by a Q-Plot column and measured by an FID. The carbon monoxide contained 5% of nitrogen, which was used as an internal standard for calculating carbon monoxide conversion. The specific selectivities were calculated on the basis of CO converted to all detected products (hydrocarbons and CO<sub>2</sub>). The carbon balance was better than 95%. The reaction was conducted at least for 48 h to reach a stable catalytic performance.

## Results and discussion

### Structure and reducibility of calcined catalysts

The structure and reducibility of the calcined catalysts were studied by XRD, Mossbauer spectroscopy and TPR combined with an *in situ* magnetic method. The XRD patterns of all calcined catalysts are shown in Fig. 1. They display peaks at 24.1°, 33.2°, 35.6°, 41.1°, 49.5°, 54.0°, 57.7°, 62.5°, and 64.1°, which could be assigned to the plane facets (012), (104) (110), (113) (024), (116) (122), (214) and (300) of Fe<sub>2</sub>O<sub>3</sub>, respectively. No other iron crystalline phases were detected by XRD. The results are consistent with previous reports<sup>15,32</sup> for silica supported iron catalysts. No diffraction peaks attributable to copper-containing or potassium-containing phases were observed. This is possibly due to low concentration of these elements in the catalysts. The diameter of the hematite crystallite size measured using the Scherrer equation is shown in Table 2. A smaller size of hematite crystallites was observed in K and Cu-promoted catalysts compared to the reference one.

The calculations show a slight increase in hematite dispersion, reflecting a reduction of the hematite particle size, in the sequence Fe/SiO<sub>2</sub> < FeCu/SiO<sub>2</sub> < FeK/SiO<sub>2</sub> < FeCuK/SiO<sub>2</sub>. Note that the iron content was the same in all these catalysts (Table 2).

The Mossbauer spectra of the calcined samples measured at high and low velocities are presented in Fig. 2 and 3, respectively. The Mossbauer parameters are summarized in

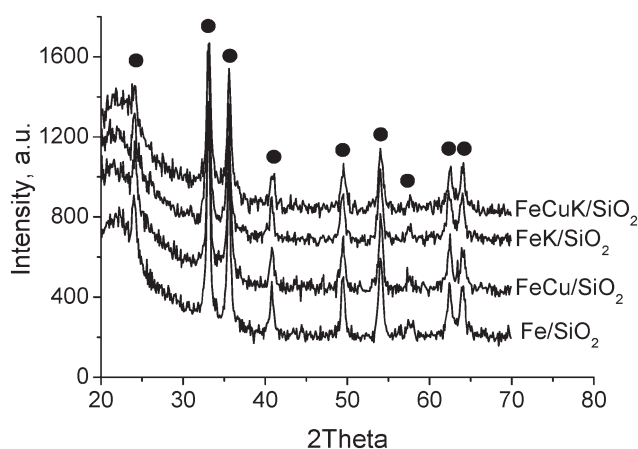


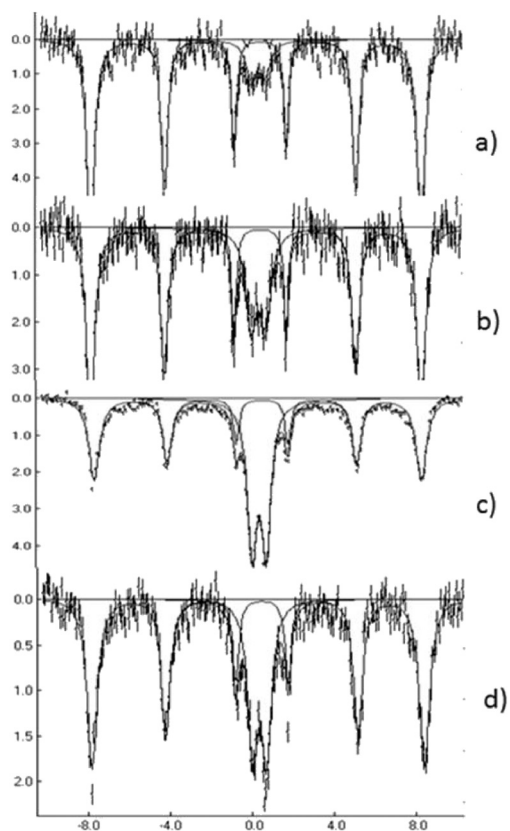
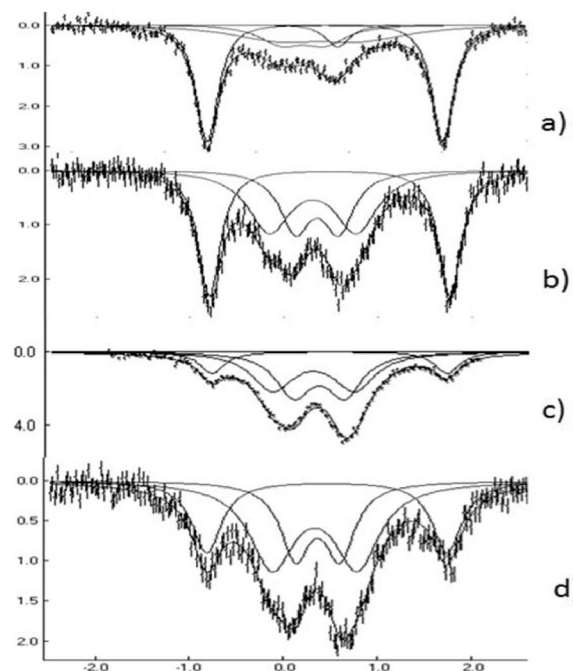
Fig. 1 XRD diffraction patterns of the calcined silica supported iron catalysts.

**Table 2** Extent of conversion of  $\text{Fe}_3\text{O}_4$  in  $\text{Fe}_5\text{C}_2$  in silica supported iron catalysts in CO and syngas

| Catalyst | Conversion of $\text{Fe}_3\text{O}_4$ to $\text{Fe}_5\text{C}_2$ , % |                        |
|----------|----------------------------------------------------------------------|------------------------|
|          | CO                                                                   | $\text{H}_2/\text{CO}$ |
| Fe       | 75                                                                   | 75                     |
| FeCu     | 63                                                                   | 77                     |
| FeK      | 63                                                                   | 71                     |
| FeCuK    | 79                                                                   | 81                     |

Table S1, ESI†. The main component of all spectra is a sextet with the magnetic field at the Fe nucleus in the range 493.7–504.7 kOe (Table S1, ESI†, Fe4 species). This field is weaker than that for the highly crystallized  $\alpha\text{-Fe}_2\text{O}_3$  (515 kOe).<sup>33</sup> This could be due to the high dispersion of the iron oxide or to its high degree of defectiveness.<sup>34</sup> Lines of paramagnetic iron species in the spectra are clearly visible at the velocity range of  $\pm 2.5 \text{ mm s}^{-1}$ . Copper ferrite ( $\text{CuFe}_2\text{O}_4$ ) and delafossite ( $\text{CuFeO}_2$ ) were not considered in the Mossbauer analysis. The fraction of delafossite should be low since the catalyst has not been exposed to a reduction agent. Copper ferrite forms at very high temperature (900 °C) which was not used in this paper for catalyst preparation.

Decomposition of spectra suggests the presence of several iron species (Table S1, ESI†). By the isomer shift value (0.29–0.37  $\text{mm s}^{-1}$ , see Table S1, ESI†) these should be related to

**Fig. 2** Mossbauer spectra of calcined iron catalysts measured at high velocity: a) Fe/SiO<sub>2</sub>, b) FeK/SiO<sub>2</sub>, c) FeCu/SiO<sub>2</sub>, and d) FeCuK/SiO<sub>2</sub>.**Fig. 3** Mossbauer spectra of iron catalysts measured at low velocity: a) Fe/SiO<sub>2</sub>, b) FeK/SiO<sub>2</sub>, c) FeCu/SiO<sub>2</sub>, and d) FeCuK/SiO<sub>2</sub>.

the  $\text{Fe}^{3+}$  ions, placed within oxygen octahedra.<sup>35</sup> The highest concentration of these iron species was found in the Cu-containing samples. Promotion with K also increases the concentration of these species but to a lesser extent. Broader widths of the lines attributed to paramagnetic iron species (0.4 to 0.7  $\text{mm s}^{-1}$ ) point to non-uniformity of each species. The species are not likely to be assigned to highly dispersed  $\alpha\text{-Fe}_2\text{O}_3$  crystallites also showing superparamagnetism. More likely they could be assigned to the mixed K-Fe or Cu-Fe oxide compounds. The known  $\text{KFeO}_2$  and  $\text{K}_{1+x}\text{Fe}_{11}\text{O}_{17}$  ( $x = 0\text{--}1$ ) compounds are formed at temperatures higher than 800 °C and contain Fe ions in tetrahedral sites.<sup>34</sup> For them, however, the isomer shift is smaller than that observed in this work. It should be noted that the fraction of  $\text{CuFeO}_2$  cannot be higher than 7% from the total area of all lines in the spectra. This is much lower than the observed values for Fe 1 and Fe 2 species (19 and 12%, respectively, see Table S1, ESI†) in FeCu/SiO<sub>2</sub>. The analysis of Mossbauer spectra may suggest the presence of Fe 3 species. Yet, their contribution is too small. The spectrum of the FeCuK/SiO<sub>2</sub> sample shows (Fig. 3) small contributions from the magnetically split lines. Because of these contributions,  $\chi^2$  is notably higher, but these lines are poorly defined.

Fig. 4 shows the TPR profiles of the iron catalysts measured simultaneously with magnetization. The appearance of the magnetization in the system coincides with at least one peak in the TPR spectrum that corresponds to the reduction of  $\text{Fe}_2\text{O}_3$  to  $\text{Fe}_3\text{O}_4$ . The peak maximum corresponds to the highest rate of the magnetization increase caused by the formation of magnetite. Iron reduction seems to occur in two distinct stages. The first stage is ascribed to the transformation of  $\text{Fe}_2\text{O}_3 \rightarrow \text{Fe}_3\text{O}_4$ , and  $\text{CuO} \rightarrow \text{Cu}$  (in the copper-promoted catalysts). The



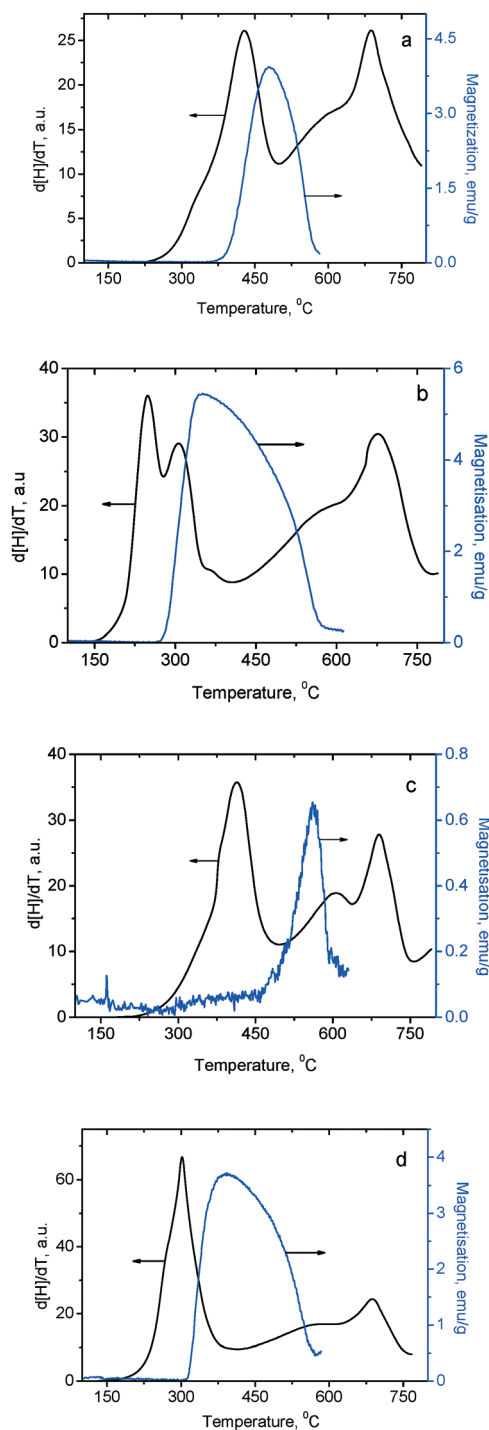


Fig. 4 Combined TPR and *in situ* magnetization profiles of silica supported iron catalysts: a) Fe/SiO<sub>2</sub>, b) FeCu/SiO<sub>2</sub>, c) FeK/SiO<sub>2</sub>, and d) FeCuK/SiO<sub>2</sub>.

following stages represent transformation of Fe<sub>3</sub>O<sub>4</sub> → FeO and then FeO → Fe. In the Cu-promoted catalysts, an additional hydrogen consumption TPR peak is observed at 220 °C which could be related to the reduction of CuO. This peak is not accompanied by any increase in magnetisation.

In our magnetisation setup, the temperature limit was 600 °C. Note that at  $T = 600$  °C when the concentration of H<sub>2</sub> was

only 5%, the reduction of magnetite to metallic iron does not occur. Under these conditions magnetite is reduced to wüstite (FeO) which is antiferromagnetic.

The TPR data clearly indicate that addition of Cu significantly promotes iron reduction for all iron reduction stages, whereas the addition of K severely hinders them. Interestingly, in the presence of copper, hematite can be reduced to magnetite at lower temperatures. The ease of iron reduction in the presence of Cu can be attributed to hydrogen dissociation on copper metallic species generated at 220 °C leading to reactive hydrogen capable of reducing iron oxides at relatively lower temperatures.<sup>36</sup> Promotion with K markedly inhibits reduction of Fe<sub>2</sub>O<sub>3</sub> to Fe<sub>3</sub>O<sub>4</sub>. The maximum magnetization corresponding to the formation of magnetite is uncovered in FeK/SiO<sub>2</sub> at temperatures 100 °C higher than that for Fe/SiO<sub>2</sub>. Such a phenomenon has been previously observed in ref. 37.

One of possible explanations may be due to the presence of a non-stoichiometric potassium ferrate KFeO<sub>x</sub> on the surface of the hematite particles during the preparation of the catalyst,<sup>38–40</sup> preventing hematite reduction. The simultaneous presence of Cu and K in the system shows a multidirectional effect on the temperature relevant to the beginning of hematite reduction.

#### Iron carbidisation kinetics under non-isothermal conditions

The iron carbidisation experiments either in pure CO or H<sub>2</sub>/CO were performed in the reactor cell of the vibrating sample magnetometer. The magnetization was measured *in situ* during the catalyst activation. The results of temperature-programmed experiments in a flow of CO and H<sub>2</sub>/CO are shown in Fig. 5a and b, respectively. The increase in the magnetization starts at 200 °C and ends at 300 °C. Then the magnetization drops to almost zero at 450–550 °C. As shown previously,<sup>15,40</sup> the magnetization growth in the presence of carbon monoxide or syngas at relatively low temperature is due to reduction of hematite (Fe<sub>2</sub>O<sub>3</sub>) to magnetite (Fe<sub>3</sub>O<sub>4</sub>). The subsequent drop in magnetization at higher temperatures is caused by the formation of the Hägg carbide ( $\chi$ -Fe<sub>5</sub>C<sub>2</sub>) with the Curie temperature of 250 °C. Therefore, above the 250 °C the catalyst magnetization decreases despite progressive formation of iron carbide. The reduction of hematite to magnetite over the unpromoted catalyst proceeds at the same temperatures in syngas and pure carbon monoxide.<sup>15</sup>

Note that iron carbidisation is affected by the presence of promoters. The copper addition leads to the decrease in the temperature of the start of both the hematite to magnetite reduction and formation of carbide both in CO and syngas. Interestingly, promotion with potassium inhibits the reduction in CO and significantly promotes it in H<sub>2</sub>/CO.

#### Iron carbidisation kinetics under isothermal conditions

The carbidisation experiments were conducted under isothermal conditions at several temperatures. We have previously shown<sup>15</sup> that the pretreatment of calcined iron catalysts first

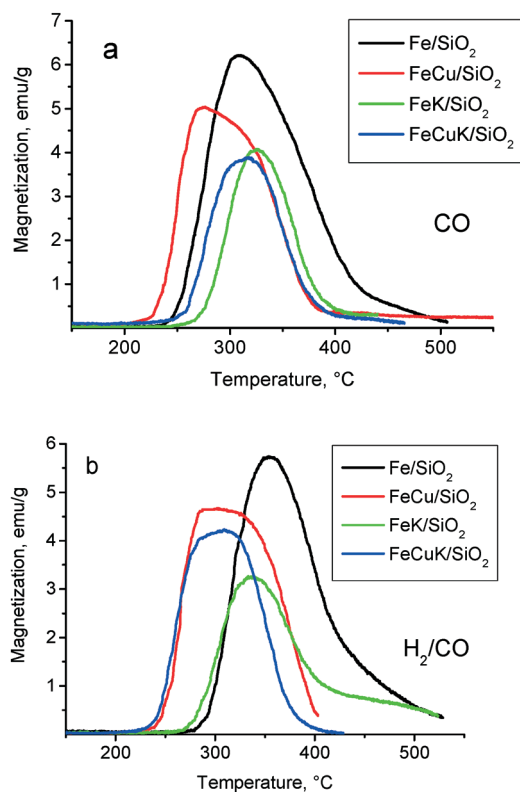


Fig. 5 Temperature dependence of the magnetization in the temperature-programmed reaction of CO (a) and H<sub>2</sub>/CO (b) with the silica supported iron catalysts.

leads to magnetite which is then converted to the  $\chi$ -Fe<sub>5</sub>C<sub>2</sub> iron carbide. This peculiarity allows separate investigation of the kinetics of Fe<sub>2</sub>O<sub>3</sub> to Fe<sub>3</sub>O<sub>4</sub> reduction and Fe<sub>3</sub>O<sub>4</sub> to  $\chi$ -Fe<sub>5</sub>C<sub>2</sub> carbidisation under isothermal conditions. In the first case, the reaction can be followed by the increase in the magnetization as the result of magnetite formation, and in the second case by the decrease in magnetization due to the formation of Hägg carbide, which is not ferromagnetic at these conditions. Note that the Curie temperature of magnetite is 570 °C,<sup>41</sup> while the Curie temperature of Hägg carbide is significantly lower (around 250 °C).

The thermomagnetic analysis of catalysts, exposed to CO at 288, 320, 350, 363 and 417 °C, or syngas at 248, 260, 283, 312, 320 and 350 °C until stabilization of the magnetization showed that the Curie temperature of the produced phase corresponds to magnetite (570 °C). The absence of inflection points in thermomagnetic curves is indicative of the presence of only the magnetite phase. No carbide forms under these conditions. Similarly, it was shown that the subsequent treatment of the catalysts preliminary reduced in carbon monoxide at 388, 410, 470, 490, 520, and 540 °C and syngas at 350, 384, 396, 410 and 440 °C leads to the formation of Hägg iron carbide phase with the Curie temperature of 250 °C. The absence of inflections in the thermomagnetic curve of the carbidized iron catalysts indicates the absence of cementite (θ-Fe<sub>3</sub>C) with a Curie temperature of 207 °C.

### Kinetics of Fe<sub>2</sub>O<sub>3</sub> reduction to Fe<sub>3</sub>O<sub>4</sub> in CO or syngas

To measure the kinetics of hematite reduction to magnetite, the catalysts were heated in an Ar flow to the desired temperature, then the Ar flow was switched to that of CO or H<sub>2</sub>/CO. The conversion was calculated in accordance with the formula:

$$\alpha = \sigma(t)/\sigma(\infty),$$

where  $\sigma(t)$  is the magnetization at time  $t$  and  $\sigma(\infty)$  is the value of the magnetization which corresponds to the complete reduction of Fe<sub>2</sub>O<sub>3</sub> to Fe<sub>3</sub>O<sub>4</sub>. An example of kinetic curves calculated from variation of magnetization at 310 °C is shown in Fig. 6.

For the interpretation of the kinetic experiments, the following kinetic expression has been used:

$$\frac{d\alpha}{dt} = k(T)f(\alpha)$$

where  $\alpha$  is the conversion,  $k$  is the rate constant at a certain temperature, and  $f(\alpha)$  is a function of the degree of transformation which reflects the mechanism of the process. The kinetic constant  $k$  is presented below:

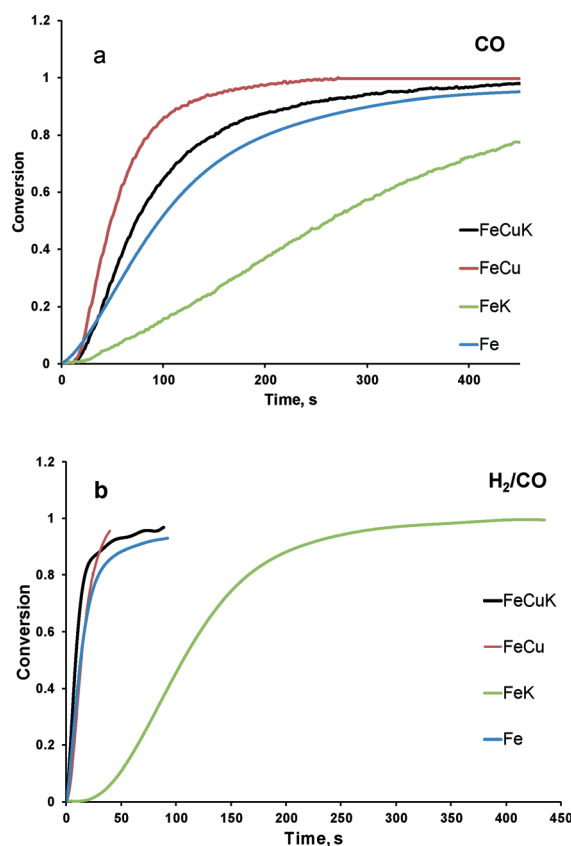


Fig. 6 Conversion of hematite into magnetite as a function of the reaction time in CO at 288 °C (a) and in syngas at 312 °C (b).

$$k = k_0 \exp\left(-\frac{E^*}{RT}\right)$$

where  $k_0$  is the pre-exponential factor,  $E^*$  is the effective activation energy,  $R$  is the gas constant, and  $T$  is the reaction temperature.

Only rarely in the gas–solid reactions,  $E^*$  does not depend on the degree of conversion, while  $k$  is a constant.<sup>42</sup> The sigmoid curve of the conversion *versus* time (Fig. 6) indicates the important role of nucleation processes in the kinetics of the  $\text{Fe}_2\text{O}_3$  reduction to  $\text{Fe}_3\text{O}_4$ . The role of the nucleation process in the reduction of hematite has been previously discussed in ref. 43. The Friedman's isoconversional method<sup>44</sup> provides an opportunity to measure the dependence of  $E^*$  on the conversion. At the same time, the experimental data can be presented as  $d\alpha/dt$  *versus*  $\alpha$  for a range of temperatures: each value of  $\alpha$  corresponds to a number of values of  $d\alpha/dt$ . Thus, for any value of  $\alpha$ , a plot of  $\ln(d\alpha/dt)$  *versus*  $1/T$  can be drawn, and both the effective activation energy and the pre-exponential factor  $k_0$  can be calculated for a given conversion. Table S2, ESI† displays the values of effective activation energies, pre-exponential factors, and calculated rate constants for reactions of hematite reduction to magnetite in CO and syngas for two values of conversion, 0.1 and 0.8. The rate constants are normalized to the specific surface area of iron oxide.

Fig. 6 and Table S2, ESI† suggest that the presence of K leads to a noticeable decrease in the rate of formation of magnetite during the hematite reduction by CO. When the  $\text{H}_2/\text{CO}$  flow was applied to  $\text{FeK}/\text{SiO}_2$ , the rate of  $\text{Fe}_3\text{O}_4$  formation was also significantly reduced. In contrast, the presence of Cu resulted in an increase in the rate of magnetite formation in both carbon monoxide and syngas. Likewise, in both atmospheres, a decrease in the rate of reduction was observed when the degree of conversion grew. There was a manifest increase in the reduction rates at both low and high degrees of conversion.

### Kinetics of $\text{Fe}_3\text{O}_4$ reduction to $\chi\text{-Fe}_5\text{C}_2$

A sharp increase in magnetization was first observed over the calcined silica supported iron catalysts when the Ar flow was switched to that of CO or  $\text{H}_2/\text{CO}$  at higher temperatures due to the initial formation of the ferrimagnetic magnetite. Then, after stabilization of magnetization, a slow decrease in the magnetization can be observed, reflecting formation of  $\chi\text{-Fe}_5\text{C}_2$ . The magnetization drops during the conversion of magnetite into carbide, as the Curie temperature of Hägg carbide  $\chi\text{-Fe}_5\text{C}_2$  is approximately  $250 \pm 5$  °C.<sup>41</sup> Assuming that only magnetite has measurable magnetization at this temperature, conversion of magnetite to  $\chi\text{-Fe}_5\text{C}_2$  can be calculated from the *in situ* magnetization data by formula:

$$\alpha(t) = \frac{\sigma_{\max} - \sigma(t)}{\sigma_{\max}} \times 100\%$$

where  $\sigma(t)$  is the magnetization at moment  $t$  and  $\sigma_{\max}$  is the maximum value of magnetization. The plot of the magnetiza-

tion from the maximum to the minimum of magnetization at temperatures higher than the Curie temperature of Hägg iron carbide reflects the kinetics of iron carbide formation. When after the carbidisation, the catalysts were cooled down to room temperature well below the Hägg carbide Curie temperature, the catalyst magnetization again increases. The amount of produced carbide was calculated by measuring magnetization at room temperature from the value of the specific Hägg carbide magnetization of  $128 \text{ emu g}^{-1}$ .<sup>41</sup> The iron carbide amount is given in Table 2. Fig. 7a and b depict the evolution of conversion of magnetite to iron carbide in the presence of CO and  $\text{H}_2/\text{CO}$  at 400 °C, respectively. Promotion of iron catalysts with potassium and copper results in a major increase in the rate of magnetite carbidisation in CO (Fig. 7a). Note that the effect of promotion on the carbidisation is more pronounced for the copper promotion compared to potassium. The higher carbidisation rate was observed for the catalysts containing both copper and potassium. A decrease in the rate of magnetite carbidisation was observed in syngas for the catalyst promoted with potassium (Fig. 7b). Table S3, ESI† shows the values of effective activation energies, pre-exponential factors, and calculated rate constants for the carbidisation of magnetite to  $\chi\text{-Fe}_5\text{C}_2$  for two values of conversion, 0.1 and 0.6. Note that at the conversion level higher than 0.6, the reaction proceeds very slowly and therefore the error in calculating  $E^*$  and  $k_0$  was higher.

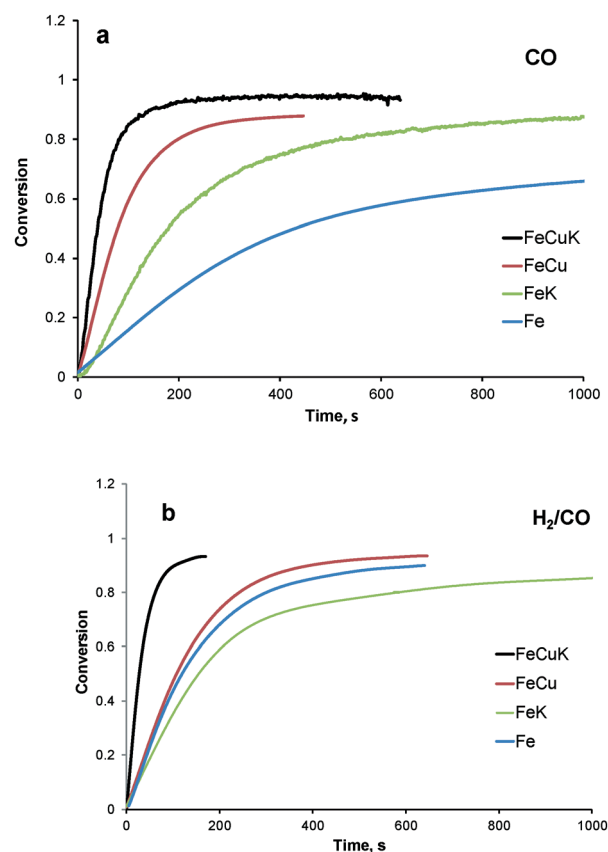


Fig. 7 Conversion as a function of the reaction time for magnetite carbidisation into Hägg carbide in CO (a) and in syngas (b) at 410 °C.

**Table 3** Catalytic performance of unpromoted and promoted silica supported iron catalysts in FT synthesis (on the basis of CO converted to all detected products including CO<sub>2</sub>)

| Catalyst               | $X_{\text{CO}}$<br>(%) | Selectivities (%) |                 |                                   |                                |                | Ole/para<br>in C <sub>2</sub> –C <sub>4</sub> |
|------------------------|------------------------|-------------------|-----------------|-----------------------------------|--------------------------------|----------------|-----------------------------------------------|
|                        |                        | CO <sub>2</sub>   | CH <sub>4</sub> | C <sub>2</sub> =–C <sub>4</sub> = | C <sub>2</sub> –C <sub>4</sub> | C <sub>5</sub> |                                               |
| Fe/SiO <sub>2</sub>    | 23.0                   | 13.2              | 28.5            | 10.1                              | 23.9                           | 24.0           | 0.42                                          |
| FeCu/SiO <sub>2</sub>  | 33.9                   | 15.9              | 20.8            | 15.2                              | 15.7                           | 32.1           | 0.97                                          |
| FeK/SiO <sub>2</sub>   | 29.9                   | 20.9              | 12.9            | 18.7                              | 6.1                            | 41.3           | 3.06                                          |
| FeCuK/SiO <sub>2</sub> | 34.3                   | 22.7              | 14.0            | 18.1                              | 7.4                            | 37.9           | 2.46                                          |

Conditions:  $T = 300\text{ }^{\circ}\text{C}$ ,  $P = 20\text{ bar}$ , GHSV =  $16\,000\text{ cm}^3\text{ g}^{-1}\text{ h}^{-1}$ ,  $\text{H}_2/\text{CO} = 2$ .

Several important observations can be deduced from Tables S2 and S3.† First, the rate of iron carbide formation during carbidisation of magnetite on all catalysts is almost two orders of magnitude lower than the rate of hematite reduction to magnetite in both CO and H<sub>2</sub>/CO. This fact allows us to discard the carbide formation when studying the kinetics of the Fe<sub>2</sub>O<sub>3</sub> reduction to Fe<sub>3</sub>O<sub>4</sub> at the initial time of the exposure of the catalysts to CO or syngas. Second, there are several remarkable effects of promotion with potassium and copper on the catalyst reduction and carbidisation. Table S2, ESI† shows that the presence of K substantially reduces both the rate of magnetite formation during hematite reduction in CO (by 7–8 times) and in syngas (by 1.5–1.3 times). We believe this is due to the formation of KFeO<sub>2</sub> or K<sub>x</sub>Fe<sub>22</sub>O<sub>34</sub> mixed compounds<sup>40</sup> during the catalyst preparation. The possibility of interactions of an alkali metal oxide with hematite during the preparation of the Fe/SiO<sub>2</sub> catalyst was previously observed.<sup>32</sup> It can be expected that in the presence of hydrogen (in the H<sub>2</sub>/CO mixture), the effect of K on the decrease in the reduction rate is lower as hydrogen apparently destroys the K–Fe–O surface compounds. Surprisingly a significant accelerating effect of potassium was observed on the carbidisation in carbon monoxide.

Tables S2 and S3† are indicative of the major enhancement of both the hematite reduction to magnetite and magnetite carbidisation to iron carbide in the presence of copper. In particular, copper produces a very significant impact on carbidisation; the carbidisation rate increases by 6 times. Note that the effect of copper on iron carbidisation is much more significant compared to that due to potassium. Carbide formation is associated with the diffusion of carbon into the crystal lattice of magnetite. One of the possible explanations of the enhancement of magnetite carbidisation in the presence of copper and to some extent of potassium can be due to the lattice defects, which are inclusions of K and Cu, and which can contribute to the diffusion of carbon atoms<sup>45</sup> during iron carbide formation.

#### Catalytic performance of unpromoted and promoted iron catalysts in high temperature FT synthesis

Methane, light paraffins, olefins and C<sub>5</sub>+ hydrocarbons, carbon dioxide and water were major products of carbon monoxide hydrogenation on iron catalysts. No noticeable deactiva-

tion was observed during the first 48 h of the catalytic test at 300 °C. The catalytic data for unpromoted and Cu- and K-promoted iron catalysts are displayed in Table 3. Promotion of the reference Fe/SiO<sub>2</sub> catalyst with either potassium or copper results in an increase in carbon monoxide conversion from 23 to 29.9% or 33.9%, respectively (GHSV =  $16\,000\text{ cm}^3\text{ g}^{-1}\text{ h}^{-1}$ ). Interestingly, the effect of copper on CO conversion and the FT reaction rate was more significant than that of potassium. Indeed, this is consistent with the higher rate of magnetite conversion to iron carbides in the Cu-promoted catalysts activated in both CO and syngas (Fig. 7) and higher iron dispersion (Table 1).

Though our results also suggest that promotion with potassium also results in higher carbon monoxide conversion, the most significant changes introduced with the addition of potassium are relevant to the hydrocarbon selectivity. Table 3 displays higher olefin and C<sub>5</sub>+ selectivities in the catalyst promoted with potassium. The olefin to paraffin ratio also increases with potassium promotion. This observation is consistent with previous reports. The effect of alkali ions is often explained by electron donation leading to the decrease in the strength of C–O bonds in carbon monoxide and a higher rate of CO dissociation.<sup>46–48</sup> It is also known that alkali metals suppress secondary olefin hydrogenation and increase the chain growth probability. Both characterization and catalytic data suggest that the effect of potassium on the catalytic performance of the promoted iron catalysts can be due both to the ease of magnetite carbidisation and higher iron dispersion resulting in a higher FT reaction rate and to the electronic interaction effects affecting the reaction selectivity. The electronic effect of potassium on iron carbide could be due to its electronic state. Recently, it was shown that maintaining copper in the +1 oxidation state during FT synthesis gives a similar effect on the hydrocarbon selectivity.<sup>49</sup> The FeCuK/SiO<sub>2</sub> catalyst promoted with both Cu and K exhibits both higher FT reaction rate and enhanced selectivity to C<sub>2</sub>–C<sub>4</sub> olefins and C<sub>5</sub>+ hydrocarbons which is due to the synergy of copper and potassium promotion.

## Conclusion

The promotion of silica supported iron catalysts with copper and potassium significantly affects reduction and carbidisation of iron species. The hematite phase was



detected in the calcined silica supported iron catalysts. Formation of the iron carbide phase in both unpromoted and Cu- and K-promoted catalysts proceeds in two steps with intermediate formation of magnetite. Addition of copper promotes both hematite reduction to magnetite and carbidisation of hematite to Hägg iron carbide. In the presence of potassium, the reduction of hematite to magnetite is somewhat slower and magnetite carbidisation in carbon monoxide is enhanced compared to the unpromoted catalyst however to a slightly lower extent than the copper promoted counterpart. Iron dispersion is enhanced in the presence of both promoters compared to the unpromoted catalyst.

A higher FT reaction rate was observed in the presence of both copper and iron promoters in the silica supported catalyst. The effect of copper on the FT reaction rate was more significant than that of potassium. Though promotion with potassium also results in a higher FT reaction rate, the more significant effect of this promoter was observed on hydrocarbon selectivity presumably due to electronic interactions between iron carbide and potassium.

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